

Total No. of Questions : 10]

SEAT No. :

PB-4966

[Total No. of Pages : 4

[6263]-627

B.E. (Mechanical)

MECHANICAL SYSTEM DESIGN

(2015 Pattern) (402048)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8 and Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) What is structural formula Explain the procedure to construct the structural diagram for machine tool gear box. Draw structural diagram for following structural formulae $3(2)2(1)$. **[6]**

b) Explain design and natural tolerances. **[4]**

OR

Q2) a) Why there is necessity of machine tool gear box for machine tool. Compare Structural Diagram & Speed Diagram. **[6]**

b) Two populations X and Y added together. Derive the expression to find mean of the resultant population. **[4]**

Q3) A belt conveyor is to be designed to carry the bulk material at the rate of 300×10^3 Kg/hr with following details, **[10]**

Density of bulk material = 800 Kg/m^3

Angle of repose of bulk material = 150°

Belt speed = 10 Km/hr

Material factor for plies, $K_1 = 2.0$

Belt tension and arc of contact factor, $K_2 = 63$

No. of plies for the belt = 4

Determine :

- a) The Suitable width of belt
- b) The Diameter and length of drive pulley

OR

P.T.O.

Q4) a) A horizontal belt conveyor is used for transporting the bulk material having mass density of 11772 N/m^3 . The surcharge factor C for the flat belt is 0.2 while the belt width is 700 mm. If the belt speed is 1.75 m/s. determine the capacity of conveyor. **[6]**

b) Explain the need of idlers in belt conveyors, also draw different types of idlers used in belt conveyors. **[4]**

Q5) a) Explain with the help of neat sketches different types of formed heads used as end closures in cylindrical pressure vessels. **[8]**

b) A seamless cylinder of storage capacity 0.025 m^3 and made of alloy steel 20Mo55 ($S_{ut} = 500 \text{ N/mm}^2$) is to be used to store a fluid at 15 MPa gauge pressure. The length of the cylinder is twice its inner diameter. If the factor of safety is 2, determine the cylinder dimensions. **[10]**

OR

Q6) a) What is auto-fretting? Describe any one method of pre-stressing the cylinders with neat sketch. **[8]**

b) A cylindrical pressure vessel with 1500mm inside diameter and 150mm thickness is subjected to internal pressure of 1.5 N/mm^2 . A nozzle of 300 mm inside diameter and 12 mm thickness is to be provided in the shell. Welds joints efficiency for shell as well as nozzle is 85 %, while corrosion allowance is 2 mm. The nozzle projects 20 mm outside and 30 mm inside the shell. If the allowable tensile stress for the shell and nozzle material is 120 N/mm^2 , determine whether the reinforcing is adequate, if not find the dimensions of reinforcing pad, if the plate thickness is 150 mm. **[10]**

Q7) a) Explain the stresses developed in the cylinder wall. **[6]**

b) Following data is given for piston of four stroke diesel engine : **[10]**

- 1) Cylinder bore = 250 mm;
- 2) Maximum gas pressure = 4 N/mm^2 ;
- 3) Bearing pressure = 15 N/mm^2 ,
- 4) Length of piston pin in bush of small end = $0.45 D$;
- 5) Ratio of inner to outer diameter of piston pin = 0.6;
- 6) Mean diameter of piston boss = $1.4 \times$ Outer diameter of piston pin;
- 7) Allowable bending stress of piston pin = 84 N/mm^2

Calculate :

- i) the outer diameter of piston pin;
- ii) the inner diameter of piston pin;
- iii) the mean diameter of piston boss
- iv) check design for bending stress

OR

Q8) The following data refers to a four stroke diesel engine : **[16]**

- a) Cylinder bore = 250 mm,
- b) Piston stroke = 300 mm,
- c) Speed = 600 rpm,
- d) Indicated mean effective pressure = 0.6 N/mm^2 ,
- e) Maximum gas pressure = 4 N/mm^2 ,
- f) Brake specific fuel consumption = 0.25 kg/kW-h
- g) Higher calorific value of fuel = $44,000 \text{ kJ/kg}$
- h) Mechanical efficiency = 80%.

Assume that 5% of total heat developed in the cylinder is transmitted by the piston. The piston is made of gray cast iron FG200 ($S_{ut} = 200 \text{ N/mm}^2$ and $K = 46.6 \text{ W/m}^2\text{C}$) and the factor of safety is 5. The temperature difference between the center and the edge of the piston head is 220°C

- i) Determine the thickness of the piston head by strength consideration and thermal consideration.
- ii) State whether a cup is required in the top of piston head, If so calculate the radius of the cup.

Q9) a) Explain the Johnsons method of optimum design. **[6]**
b) A tensile bar of length 400 mm is subjected to constant tensile force of 4000 N. If the factor of safety is 2 design the bar with the objective of minimizing the material cost out of the following materials. **[10]**

Material	Mass density, ρ (Kg/m^3)	Material Cost, C (Rs/N)	Yield Strength S_{yt} (MPa)
Plain carbon steel	7800	28	400
Alloy steel	7850	150	900
Al alloy	2800	140	150
Titanium alloy	4500	2200	800

OR

Q10) A shaft is to be used to transmit a torque of 900 N-m. The required torsional stiffness is 90 N-m /degree while the factor of safety based on yield strength is 1.5. Using maximum shear stress theory, design the shaft with the objective of minimizing the weight. What will be the change in design for minimum cost? **[16]**

Material	Mass Density p. kg/mm ³	Modulus of rigidity G, (GPa)	Yield strength Syt, MPa	Material Cost, C (Rs/N)
MAT1	8500	80	130	16
MAT2	3000	26.5	50	32
MAT3	4800	40	90	480
MAT4	2100	16	20	32
